Ruhan Bhavsar

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechatronics Engineering — *Expected 2028*

Experience

Embedded Systems Validation Developer (Motor Control) — Sept 2026 - Dec 2026
onsemi — *Waterloo, ON*

Sales Automation & Engineering Support Intern — Jan 2026 - Apr 2026
MPI Morheat — *Toronto, ON*

- Developed a **React-based** configurator for flange, screwplug, and PTFE heater drawings, reducing custom drawing generation time by **95%** from approximately **20 minutes to 1 minute** for customer proposals.
- Developed a low-cost (<**$500**) automated **three-axis label cutter** using **stepper motors**, **3D-printed components**, and an **STM32 microcontroller** programmed in **C++**, **saving 5–10 hours of manual labor per week**.

Engineering Intern — May 2025 - Aug 2025
Vexos — *Markham, ON*

- Built manufacturing tools with **Python**, **SolidWorks**, and **3D printing** that reduced equipment downtime by **40 hours monthly**.
- Created standardized instructions that cut operator training time by **2 hours** and reduced assembly-line errors by **25**%.

Firmware Developer — Sept 2024 - Present
Midnight Sun Solar Car Team — *Waterloo, ON*

- Maintained embedded software for **8 microcontrollers**, managing critical **real-time tasks** and **ADC sampling** with a **10 ms latency target** to ensure system-wide synchronization.
- Implemented a **Hardware-in-the-Loop (HIL)** testing framework using **Pytest** and **Python scripts**, reducing manual testing time by **80**%.

Projects

Natural Disaster Severity Mapping | *React, Flask, MongoDB, PyTorch, ResNet34*

- Built a full-stack web application using **React**, **Flask**, and **MongoDB** to assess earthquake severity from user-uploaded images.
- Trained a **ResNet34 CNN** in **PyTorch** to classify disaster severity with **95% accuracy** across **8,500 images**.
- Integrated the **Mapbox API**, secure authentication, and real-time map updates into a responsive user interface.

Anchor - App Blocker | *Kotlin, Jetpack Compose, Android SDK, NFC*

- Developed an end-to-end native **Android application** that pairs with any **NFC card** to activate and deactivate app-blocking sessions.
- Implemented configurable **app restrictions**, daily usage limits, and recurring schedules that automatically enforce users' focus routines.
- Built goal and streak tracking with **persistent progress data**, creating a complete digital-wellbeing experience designed for repeated daily use.

Arduino Emergency Responder | *React, Flask, Python, C++, PySerial*

- Developed a full-stack emergency-response application with a **React** frontend and **Flask** backend, using **PySerial** to process real-time alerts from an Arduino.
- Designed a severity-based prioritization algorithm and interactive dashboard to rank incidents and display live emergency updates.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, Kotlin, C++, C, SQL, HTML/CSS
Frameworks/Libraries: React, Flask, Node.js, Jetpack Compose, Tailwind CSS, PyTorch, Pytest
Databases/Tools: MongoDB, Git, Linux, Docker, Android Studio, Vagrant, SCons
Concepts: REST APIs, Full-Stack Development, Machine Learning, Data Structures & Algorithms, Design Patterns, Unit Testing, CI/CD